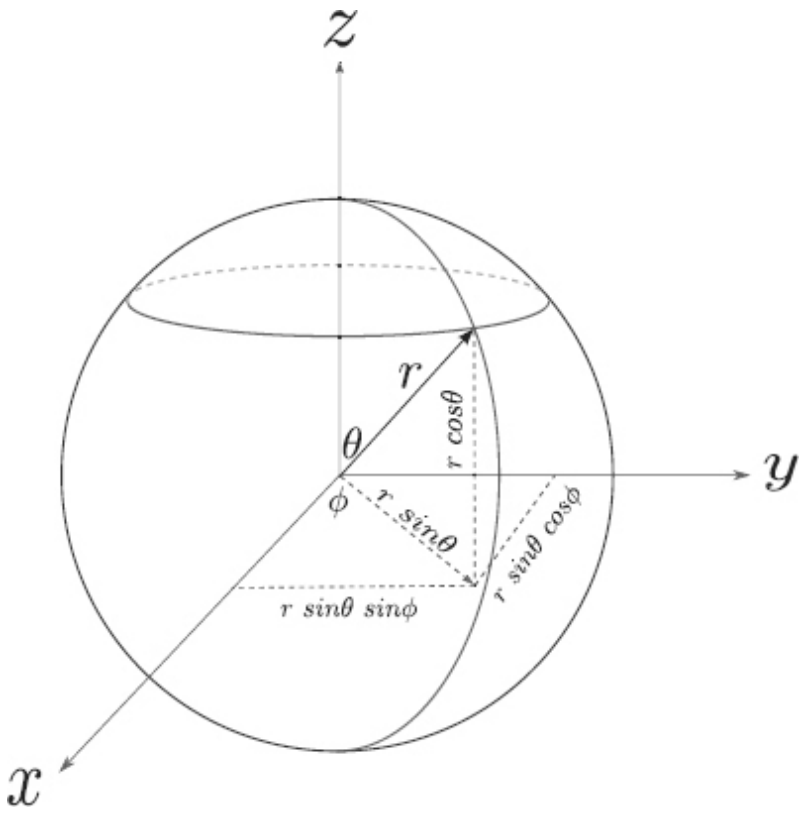


Figure 3.2: Spherical Coordinates. This diagram illustrates conventional spherical coordinate labels r, θ, and φ. It also illustrates the conversion to Cartesian coordinates: $x = r \sin\theta \cos\varphi$, $y = r \sin\theta \sin\varphi$, and $z = r \cos\theta$.

You could also try working out a much more elaborate example involving two directions, $\hat{n}$ and $\hat{m}$. In this setup, $\mathcal{A}$ not only *ends up* in an arbitrary direction; it also *starts out* in a (different) arbitrary direction.

Exercise 3.5: Suppose that a spin is prepared so that $\sigma_m = +1$. The apparatus is then rotated to the $\hat{n}$ direction and σ_n is measured. What is the probability that the result is +1? Note that $\sigma_m = \sigma \cdot \hat{m}$, using the same convention we used for σ_n.

The answer is the square of the cosine of half the angle between $\hat{m}$ and $\hat{n}$. Can you show it?